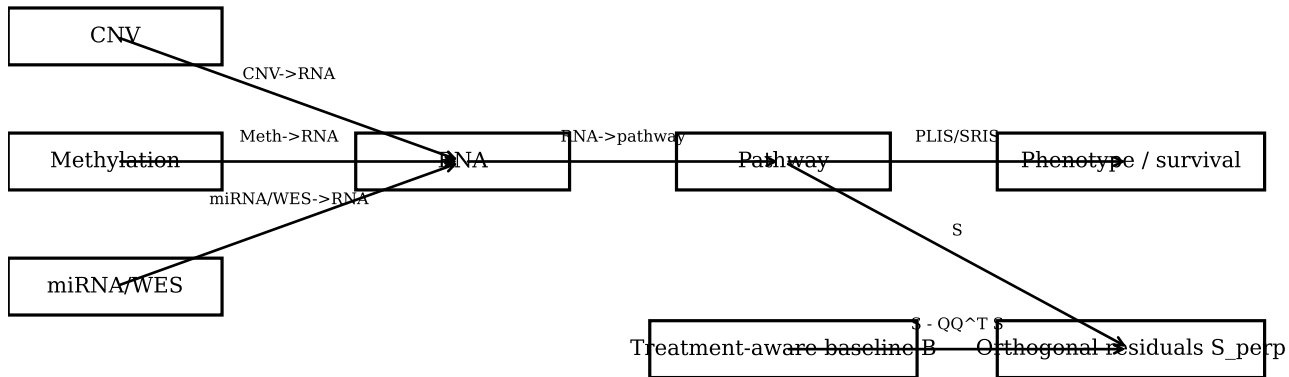


Multi-omics cellular sheaf architecture

$$r_e^\theta(p) = \rho_{u,e}^\theta(x_u(p)) - x_v(p), \quad \rho^\theta(x) = Ax + U\sigma(Vx + b)$$



Treatment-aware QR residualization: $S_{\perp} = S - QQ^{\top} S$, with $\text{span}(Q) = \text{col}(B)$